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Sediment-resistant marine antifouling amphiphilic random copolymer coating

Jinhyeok Jeong¹, Minjoong Shin², Sung Min Kang¹ and Myungeun Seo^{2,3}

Abstract

Accumulation of aquatic organisms on submerged sea vessel surfaces increases drag, resulting in higher fuel consumption and more greenhouse gas emissions. Polymer coatings hold promise to combat marine biofouling as more environmentally friendly alternatives to metal-based paints. While hydrophilic coatings are known to lose antifouling efficacy upon sediment adsorption, how to arrange hydrophobic segments and develop a sufficiently thick coating layer to suppress diatom deposition in the presence of silt and maximize the antifouling performance has been unknown. Here, we developed an effective and scalable antifouling coating by combining the polydopamine precoat method with well-defined amphiphilic copolymers synthesized using a controlled polymerization technique. Reversible addition-fragmentation chain transfer copolymerization of zwitterion-containing hydrophilic sulfobetaine methacrylate and hydrophobic trifluoroethyl methacrylate produced the target copolymers with control of composition, molecular weight, and sequence. Zr(IV)-mediated coordination bonds between polydopamine and sulfobetaine groups yielded >20 nm thick films stable for a month in seawater. The random copolymer sequence exposed both hydrophilic and hydrophobic groups on the outermost coating surface. Synergistic repelling diatom and silt led to the best antifouling performance at the optimal hydrophilic-hydrophobic balance. Superior antifouling efficacy was retained for a month and was also effective on stainless steel, suggesting the potential for practical application.

Introduction

Amphiphilic copolymer coatings have garnered significant interest as environmentally friendly and sustainable marine antifouling coatings to mitigate the toxicity of currently used biocide-containing materials^{1–10}. This interest stems from the observed shortcomings of previously developed hydrophilic polymer coatings, which demonstrate poor antifouling performance in the presence of marine sediments (Fig. 1a)^{11,12}. Hydrophilic polymer coatings form a hydration layer on solid surfaces that acts as a physical barrier against fouling organisms such as diatoms in sediment-free conditions^{13–18}. However, they lose the fouling resistance upon sediment

adsorption^{11,19}. As integrating a hydrophobic character into the polymer coating can be effective in repelling marine sediments (silt), amphiphilic copolymer coatings possessing both hydrophilic and hydrophobic properties can provide a synergistic platform to combat marine biofouling in practical conditions¹². Particularly, combining hydrophobic segments with zwitterionic species as a hydrophilic component has attracted recent attention due to strongly immobilized hydration layer formation^{5,12,20–31}. A few studies demonstrated the potential of zwitterion-containing amphiphilic coatings in the ocean field assay^{5,12,20,28}. Surprisingly, however, there has been no report that evaluated the diatom deposition on the copolymer surface in the presence of silt.

Despite the successful application of these copolymers in marine antifouling coatings, several challenges remain unmet^{6,32–35}. The coating process should be able to fix the copolymer onto materials covering the hull exterior, such as steel and fiberglass, on a large area. While simple polymer coating relying on physical adhesion does not

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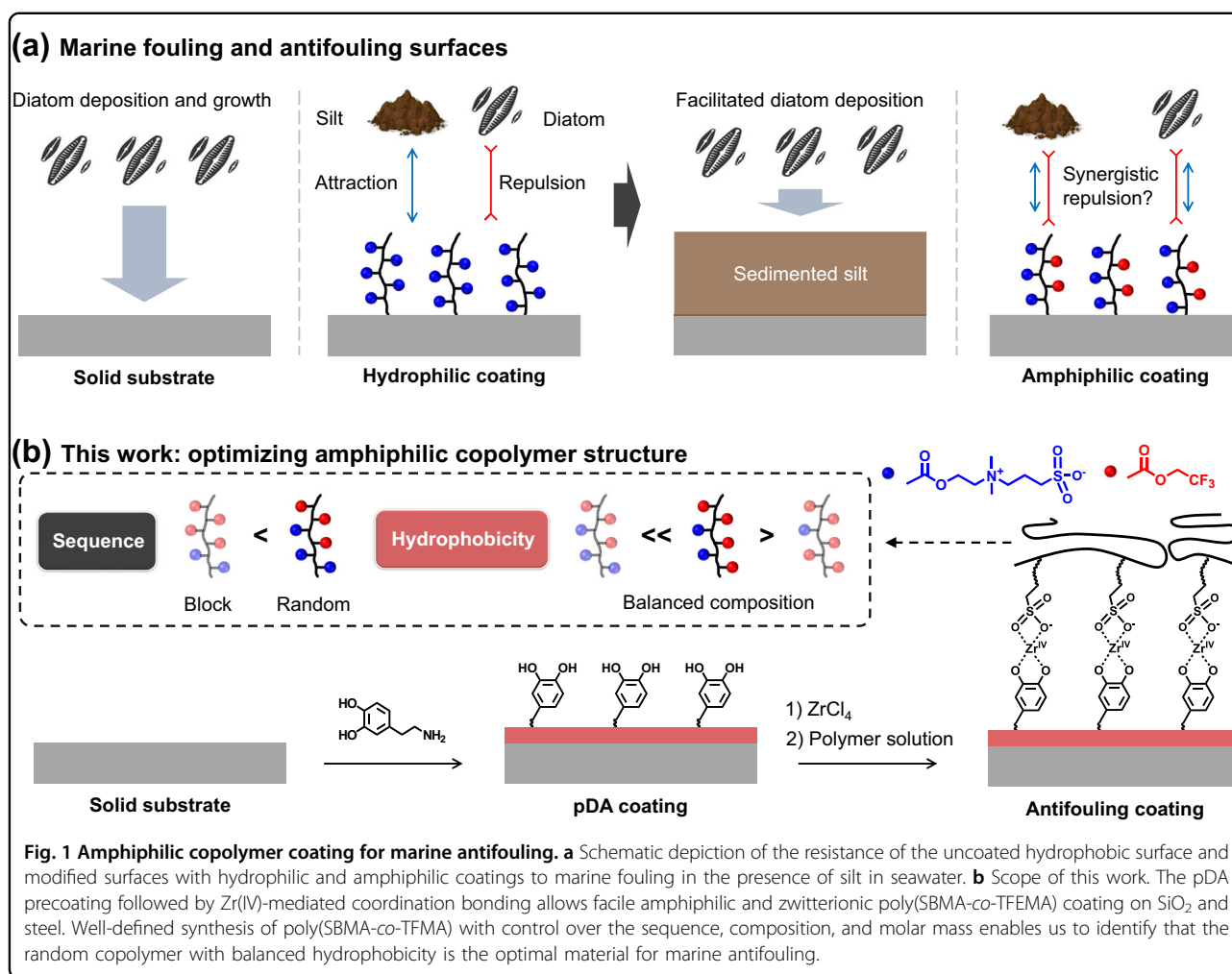
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provide a strong union to the surface, additional and surface-specific treatments involving chemical reactions are required to covalently graft the copolymers to the surface, making the approaches less appealing for the application^{4,5}. Being typically synthesized by copolymerization of hydrophilic and hydrophobic monomers, the resulting amphiphilic copolymer can possess different average compositions with diverse sequence distributions, ranging from an ensemble of random binary sequences to diblock copolymer chains where a single junction joins two individual blocks comprising each monomer³⁶. However, the effects of structural parameters on the fouling resistance remain unclear. To address these challenges, a universal and scalable grafting methodology needs to be developed and combined with amphiphilic copolymers. As surface grafting via in situ polymerization typically does not allow thorough solution-based characterization of the resulting polymers, well-defined amphiphilic copolymers should be separately synthesized with control of molar mass, composition, and sequence distribution and fully characterized before

grafting. The systematic investigation on the influence of the structural parameters will allow us to establish a design principle for marine antifouling materials.

In this study, we develop a polydopamine (pDA) precoating-mediated methodology for grafting of amphiphilic zwitterionic copolymers. The mussel-inspired pDA coating facilitates ad-layer formation in a simple step and substrate-independent manner, exploiting the unique catechol-surface interaction and the spontaneous oxidative polymerization^{37–39}. Based on our recent result, we choose 3-(*N*-2-methacryloyloxyethyl-*N,N*-dimethyl)ammonatopropanesulfonate (SBMA) and 2,2,2-trifluoroethyl methacrylate (TFEMA) as zwitterionic sulfobetaine and hydrophobic monomers, which combination in the surface-initiated grafting copolymerization showed compelling antifouling performances⁴⁰. Among various hydrophobic monomers, TFEMA was selected for its unique chemical and physical properties. For example, the C–F bond is one of the strongest, making TFEMA-containing polymers highly resistant to oxidation and heat. Additionally, TFEMA-based polymer coatings are known to effectively lower surface

energy, making them useful for non-stick applications. We note that SBMA^{5,20,25–27,31} and TFEMA^{20,21} have also been combined with other counterparts to produce amphiphilic antifouling surfaces, respectively. In contrast to our previous approach, where the initiating site for the polymerization had to be immobilized onto the surface and only thin copolymer brushes were produced (<10 nm thick)⁴⁰, we show that multiple metal ion-mediated coordination bond formation between the remaining catechol groups on the pDA layer and the sulfonate groups in poly(SBMA-*co*-TFEMA) allows for straightforward grafting of the pre-synthesized target copolymer. Sequential application of pDA, poly(SBMA-*co*-TFEMA), and ZrCl₄ solutions results in a strong and thick (~40 nm) copolymer coating with mechanical integrity, reinforced by the Zr(IV)-mediated coordination bonds, stable in seawater for a month (Fig. 1b).

To obtain well-defined poly(SBMA-*co*-TFEMA), we employ the reversible addition-fragmentation chain transfer (RAFT) polymerization as a controlled radical polymerization with high functional group tolerance to produce various poly(SBMA-*co*-TFEMA)s with controlled molar masses and TFEMA-to-SBMA ratios. The copolymerization kinetics and the chain extension experiments support random and block poly(SBMA-*co*-TFEMA)s can be successfully synthesized. While both the molar mass and the composition of poly(SBMA-*co*-TFEMA) affect the coating thickness, the sequence effect was negligible. In contrast, the much higher water contact angle of the random copolymer coating compared to block indicates a larger population of the hydrophobic TFEMA repeating units on the surface. For the first time, we report composition-dependent antifouling performances of zwitterion-containing amphiphilic copolymer coatings on Si/SiO₂ and practically more relevant stainless steel substrates against diatom in the presence of silt. In the optimized condition, random copolymer remarkably suppressed diatom adhesion by <5% compared to the uncoated Si/SiO₂ after 24 h, outperforming literature examples evaluated in sediment-free conditions. The coating was still strongly repelling after 4 weeks and was also effective on stainless steel. While the antifouling performance of the block analog was moderate, the hydrophilic poly(SBMA) homopolymer control showed even worse resistance to fouling. Our findings suggest that the balanced incorporation of the hydrophobic segments in the hydration layer is crucial for suppressing sedimentation while retaining resistance to organism deposition.

Results and discussion

Amphiphilic copolymer synthesis

The synthetic route to random copolymer, poly(SBMA-*r*-TFEMA), via RAFT copolymerization is given in Fig. 2a. We chose dimethyl sulfoxide (DMSO) as the reaction

solvent and added lithium chloride (LiCl, 13 wt% relative to SBMA) to improve the solubility of SBMA and make a homogeneous polymerization mixture. 4-Cyano-4-(dodecylsulfanylthiocarbonyl)sulfanyl pentanoic acid (CDPA) was used as the chain transfer agent to control methacrylate polymerization initiated by 2,2'-azobis(2-methylpropionitrile) (AIBN).

We investigated the copolymerization kinetics of the polymerization mixture containing [SBMA]:[TFEMA] = 1:3. While the feed composition is asymmetric, both monomers showed similar conversion over time and reached >80% conversion after 8 h, suggesting the random copolymerization behavior (Fig. 2b). ¹H NMR spectra of the resulting copolymers supported the successful incorporation of both monomers with similar compositions estimated by conversion (Supplementary Figs. S1–S5). The first-order kinetics plot showed a linear dependence consistent with the RAFT mechanism up to 8 h (Fig. 2c). The slight reduction in the slope in the later stage of polymerization was attributed to the decreased radical species by termination. Because of the drastic solubility difference between SBMA and TFEMA, estimating the molar mass of the amphiphilic copolymers by gel permeation chromatography (GPC) was challenging. We found that 1,1,1,3,3,3-hexafluoro-2-propanol (HFIP) was the best eluent that seems to reduce the interaction between the copolymer and the column yet the traces were fairly broad, giving dispersities of 1.4–2.0. Nonetheless, a gradual shift of the GPC traces to lower retention time was consistent with the increasing molar mass as a function of time (Fig. 2d). While full conversion could be achieved after 120 h, the increase in molar mass was relatively small. We synthesized a higher-molar mass copolymer by doubling the [monomer]:[CDPA] ratio from 124:1 to 248:1. Supplementary Table S1 summarizes the details of the copolymers synthesized in this study. Henceforth, we use the theoretical molecular weight ($M_{n,theo}$) estimated from conversion to represent the molecular weight of the copolymers because of the issues with molecular weight determination by GPC.

We synthesized poly(SBMA-*r*-TFEMA)s with other feed compositions to investigate the composition effect on coating efficacy and marine antifouling performance. While the polymerizations were quenched before complete conversion, the copolymer composition was again comparable with the feed, supporting random copolymerization (Supplementary Figs. S6 and S7). We designate poly(SBMA-*r*-TFEMA) as polySrT(x , y), where x and y represent the mole fraction of SBMA in the copolymer and its $M_{n,theo}$ (kg mol^{−1}), respectively. We also synthesized a block copolymer, poly(SBMA-*b*-TFEMA) (polySbT), to examine the influence of the sequence (Fig. 2a). We first obtained poly(TFEMA) by RAFT polymerization and then performed SBMA polymerization

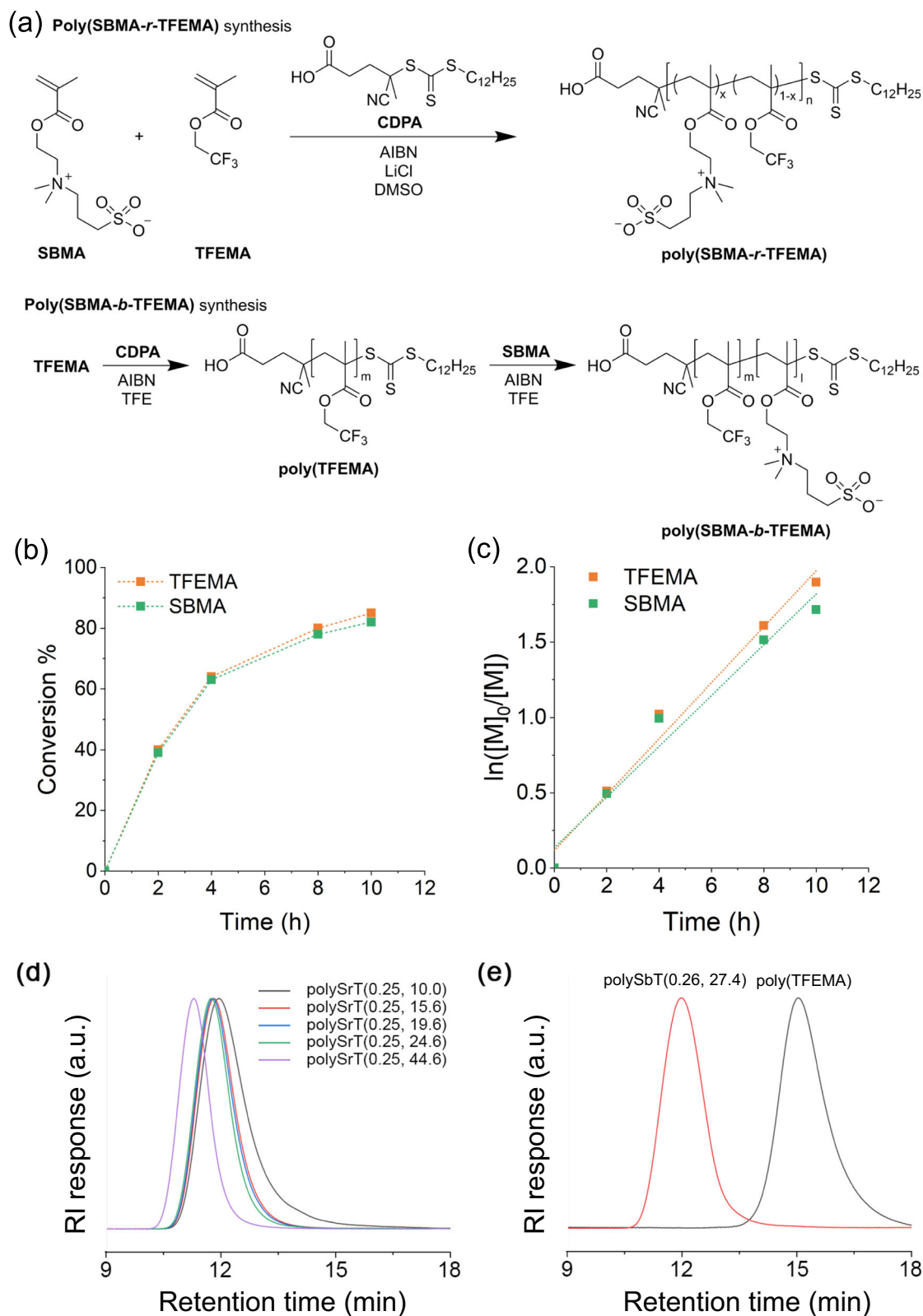
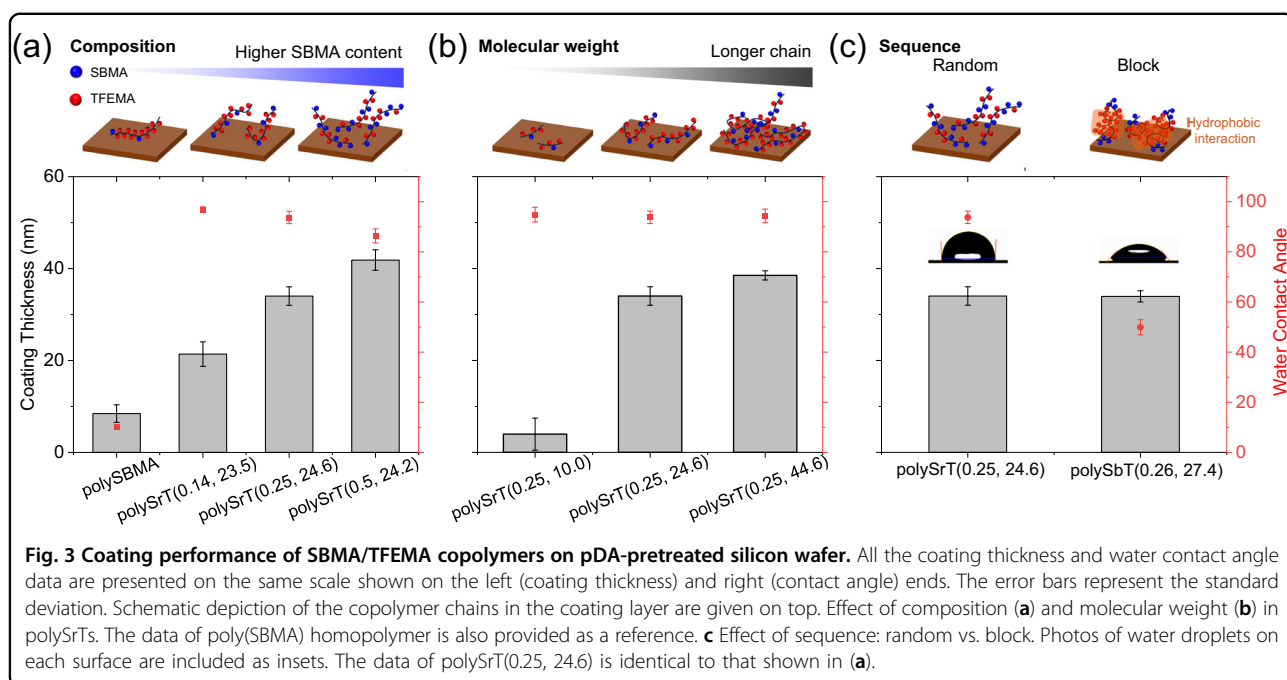


Fig. 2 Synthesis of SBMA/TFEMA copolymers. **a** Synthetic routes to the random poly(SBMA-*r*-TFEMA) (polySrT) and block poly(SBMA-*b*-TFEMA) (polySbT). **b, c** Copolymerization kinetics of TFEMA and SBMA for the random polySrT synthesis. **b** Time vs. conversion. **c** First-order kinetic plot. **d, e** SEC traces of polySrTs (**d**) and polySbT (**e**). The trace of poly(TFEMA) is shown as a reference.



using the poly(TFEMA) containing the trithiocarbonate motif at the chain end as a macro-chain transfer agent to extend the chain with poly(SBMA). A complete shift in the HFIP-GPC traces after the chain extension indicates that the target diblock copolymer composed of TFEMA and SBMA blocks was successfully attained (Fig. 2e). We adjusted the polymerization condition to produce the block copolymer with the [SBMA]:[TFEMA] ratio of 1:3 to compare its coating and antifouling properties to the random version with the comparable composition (Supplementary Fig. S8). Finally, we synthesized SBMA homopolymer (poly(SBMA)) via RAFT polymerization as a control (Supplementary Scheme S1 and Supplementary Fig. S9). M_n and dispersity ($\bar{D}$) of poly(SBMA) were estimated as 32.8 kg mol^{-1} and 1.52 by GPC.

Coating performance

The synthesized polymers were coated on the pDA-pretreated Si/SiO₂ substrate. We estimated the polymer coating thickness by spectroscopic ellipsometry (Fig. 3 and Supplementary Table S2). While poly(SBMA) homopolymer coating was <10 nm thick, incorporating the hydrophobic TFEMA repeating unit boosted the thickness above 20 nm. In the random copolymer series, the coating became thicker with the increasing SBMA content and reached >40 nm when the [SBMA]:[TFEMA] ratio was 1:1 (Fig. 3a). The SBMA content dependence supports the idea that the sulfobetaine group is responsible for pDA-surface anchoring via Zr(IV)-mediated interaction. The presence of the TFEMA repeating unit seems equally crucial for realizing thick polymer coatings.

We posit that hydrophobic interaction between the trifluoroethyl groups within the coating layer mediates successive deposition of the copolymer chains, enabling layer thickening above 10 nm. Copolymers with $M_{n,theo} > 20 \text{ kg mol}^{-1}$ produced a stable thick coating at the SBMA mole fraction of 25% (Fig. 3b). Doubling the molecular weight raised the thickness by <20%. A block copolymer analog with a comparable composition and molecular weight to the random copolymer yielded a similar coating thickness, supporting the role of the hydrophobic interaction between TFEMA repeating units (Fig. 3c).

While the poly(SBMA) coating exhibited a water contact angle < 10° due to the excellent water-binding capability of SBMA, the random copolymer coatings were much more hydrophobic, showing 86 - 97° of the contact angles (Fig. 3a and Supplementary Table S2). The water contact angle decreased with the increasing SBMA content, indicating that the surface population of sulfobetaine and trifluoroethyl groups is related to the copolymer composition. We evaluated the surface chemical composition of the copolymer coatings using X-ray photoelectron spectroscopy (XPS) analysis. The N 1s peak appeared upon pDA pretreatment and the peaks corresponding to Zr 3d, S 2p, and F 1s emerged after the copolymer coating, consistent with the coating process (Supplementary Fig. S10). The [SBMA]:[TFEMA] ratio on the surface calculated from the atomic composition was in good agreement with the copolymer composition (Supplementary Table S3). Given that TFEMA is the major component in the random copolymer within the composition range screened in this study, the results

suggest that at least some trifluoroethyl groups remain on the outermost surface even in contact with water to render the surface hydrophobic (i.e., the minor sulfobetaine units cannot cover the whole surface). We note that the change in contact angle of polySrT(0.25, 24.6) with the increasing immersion time in water was not significant.

When the copolymer composition was fixed, the contact angle showed negligible changes by varying molecular weight (Fig. 3b). In contrast, the block copolymer coating showed a much lower contact angle of 50° (Fig. 3c). The marked dependence of the wettability on the copolymer sequence is attributed to the more effective surface coverage by the pure SBMA block that does not contain TFEMA repeating units. XPS analysis indicated a much smaller TFEMA fraction on the block copolymer coating surface than random, supporting the interpretation (Supplementary Fig. S10 and Supplementary Table S3). The different chain packing also affected the coating surface roughness, as determined by atomic force microscopy (AFM). Compared to the smooth surface morphology of the random copolymer coating with a low root-mean-square (RMS) roughness of 11.2 nm, several large aggregates were observed on the block copolymer-coated surface with the RMS roughness of 62.9 nm, presumably because of association between the hydrophobic TFEMA blocks (Supplementary Fig. S11). The other random copolymers also produced uniform coatings with the RMS roughness of 8.5 nm for polySrT(0.5, 24.2) and 6.2 nm for polySrT(0.14, 23.5). Overall, the data indicates that the combination of SBMA with TFEMA in the random sequence allows us to reliably produce thick and smooth hydrophobic coatings above a critical molecular weight ($\sim 20 \text{ kg mol}^{-1}$), with control over hydrophilicity based on the copolymer composition.

We investigated the marine antifouling properties of the copolymer coatings. To mimic the natural environments where marine fouling occurs, we performed the antifouling assays under dynamic conditions, wherein polymer-coated surfaces were incubated in a suspension of marine diatoms and sediments (silt) in artificial seawater. Under such conditions, marine sediments tend to adsorb onto hydrophilic homopolymer coatings, thereby causing the coatings to fail to resist diatom adhesion. Nonetheless, the random polySrT(0.25, 24.6) coating dramatically suppressed diatom adhesion compared to the uncoated Si/SiO₂ substrate (Fig. 4a). In contrast, the hydrophilic poly(SBMA) homopolymer coating facilitated diatom adhesion even more than the bare surface (Fig. 4b). The superior antifouling performance was also observed on the practically more relevant stainless steel substrate, a primary element of marine vessels (wettability and surface chemical compositions were similar to coatings on Si/SiO₂; see Supplementary Fig. S12 and Supplementary Tables S2 and S4). Quantitatively, the number of

attached diatoms was reduced by >90% on the random copolymer coating compared to the uncoated counterparts, regardless of the underlying substrates (Fig. 4a, c). Interestingly, the block polySbT(0.26, 27.4) coating was less effective than the random copolymer despite a comparable composition and molecular weight. These findings underscore the importance of copolymer sequence on marine antifouling properties and highlight the synergy between randomly arranged hydrophilic and hydrophobic repeating units.

We systematically evaluated the marine antifouling properties of the random copolymer coatings on Si/SiO₂ as functions of composition and molecular weight (Fig. 4b). The antifouling efficiency initially increased with the increasing SBMA content from 14 to 25 mol%, then dropped significantly at 50 mol% to a level similar to the bare surface. The [SBMA]:[TFEMA] = 1:3 ratio in the random copolymer coating seems to provide optimal hydrophilicity-hydrophobicity balance that synergistically suppresses diatom adhesion and silt sedimentation. The smooth surface of random copolymer coatings with similar RMS roughness (Supplementary Fig. S11) suggests that differences in marine antifouling properties among these coatings are primarily due to variations in their chemical composition rather than surface roughness. To investigate the origin of this observation, we conducted additional adsorption experiments using silt alone. The results showed severe silt adsorption on the polySBMA and polySrT(0.5, 24.2) coatings, whereas both polySrT(0.25, 24.6) and polySrT(0.14, 23.5) coatings exhibited strong anti-silt adsorption properties (Supplementary Fig. S13). Given that the adsorption of marine diatoms tends to increase with a higher proportion of TFEMA, it is likely that the polySrT(0.25, 24.6) coating, which contains a lower proportion of TFEMA than the polySrT(0.14, 23.5) coating, exhibited the best marine antifouling properties. The outperformance of the random copolymer over block can also be understood by the presence of both sulfobetaine and trifluoroethyl groups on the outermost surface, compared to the high sulfobetaine population in the block copolymer coating. Conversely, polySrT(0.14, 23.5) may be too hydrophobic to minimize diatom adhesion. We also note that the relatively thin coating thickness of polySrT(0.14, 23.5), presumably because of the low SBMA mole fraction, could also negatively impact antifouling performance. At the optimal SBMA content of 25 mol%, the 4-nm-thick polySrT(0.25, 10.0) coating was ineffective for antifouling. Raising the molecular weight from 24.6 to 44.6 kg mol⁻¹ did not improve the performance noticeably, consistent with the similar thickness of both coatings.

To evaluate the long-term stability of marine antifouling coatings, we examined the thickness change of the polySrT(0.25, 24.6) coatings on Si/SiO₂ after incubation in

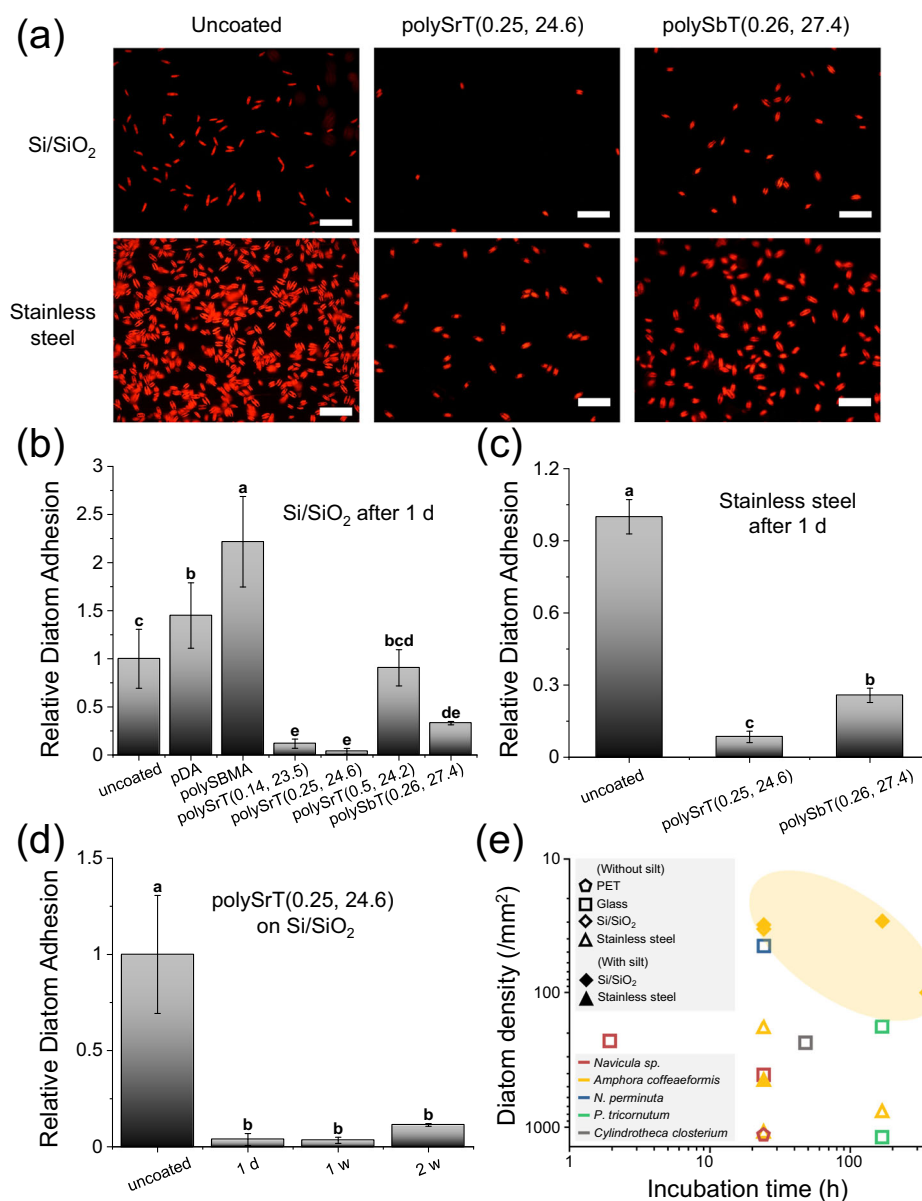


Fig. 4 Antifouling performance against diatom adhesion under seawater conditions in the presence of silt. **a** Fluorescence images of Si/SiO₂ and stainless steel substrates and substrates coated with polySrT(0.25, 24.6) and polySbT(0.26, 27.4) after exposure to diatoms for 1 d. The uncoated substrates are shown as references. The scale bars represent 50 μm. **b** Quantifying the effect of composition, molecular weight, and sequence of the copolymer coatings on the antifouling performance. The number of diatoms adhered onto the coating surface on the Si/SiO₂ substrate after 1 d was counted and normalized by the number on the uncoated substrate. **c** Antifouling performance of polySrT(0.25, 24.6) and polySbT(0.26, 27.4) coatings on stainless steel substrates. **d** Long-term stability of polySrT(0.25, 24.6) coatings on Si/SiO₂ under seawater conditions for up to two weeks. Each value is the mean of 60 counts of at least three replicate samples. The error bars represent the standard deviation. Surfaces with different labels show significant differences. **e** Diatom adhesion density on amphiphilic polymer coatings reported in the literature and developed in this study as a function of incubation time. The shade includes the data points shown in (d). See Supplementary Table S5 for more information.

artificial seawater for up to four weeks. As shown in Supplementary Fig. S14, the coating thickness remained stable over time with no significant loss. We also assessed the change in fouling resistance of the polySrT(0.25, 24.6) coating. The polySrT(0.25, 24.6) coatings that had been immersed in artificial seawater for four weeks were taken

out, and a diatom adhesion test was conducted. The number of attached diatoms did not significantly increase compared to the as-prepared polySrT(0.25, 24.6) coatings, with fouling levels remaining below 5% relative to the uncoated Si/SiO₂ substrate. This indicates that the Zr(IV)-mediated coordinative bonds hold the copolymer

coating tightly even under high-salt conditions (Supplementary Fig. S15).

We compared the antifouling performance of the polySrT(0.25, 24.6) coating against diatoms to zwitterion-containing amphiphilic copolymer coatings reported in the literature (Fig. 4e and Supplementary Table S5)^{20–22,24,25,28,29,31,40}. We note that all the data except for this study have been reported in sediment-free conditions. While direct comparison would be difficult as diatom deposition is influenced by factors including its kind and the substrate for coating, it is clear that the optimized combination of SBMA with TFEMA suppresses the adhesion of diatoms very effectively when silt was not included (34 mm^{-2} on Si/SiO₂ after 1 day). More importantly, even in the presence of silt, the developed coating retains superior performance compared to other literature examples for a prolonged period, as shown in Fig. 4d (33 mm^{-2} after 1 day and 100 mm^{-2} after 2 weeks). While this study focuses on understanding the effect of the copolymer sequence and composition on the marine antifouling coating, we anticipate that there will be more room for further performance improvement by incorporating other comonomers²⁰ and engineering the film morphology²⁷.

Discussion

We developed an amphiphilic copolymer-based new marine antifouling coating consisting of zwitterionic sulfobetaine and hydrophobic trifluoroethyl pendants. We synthesized the copolymers by controlled radical copolymerization of SBMA and TFEMA following the RAFT mechanism with control of composition, molecular weight, and sequence. Zr(IV)-mediated coordinative bonding between the sulfobetaine group in the copolymer and the pDA-pretreated surface readily produced >30-nm thick coatings when the copolymer molecular weight exceeded 20 kg mol^{-1} . Well-defined characteristics of the copolymers allowed us to systematically investigate the effect of the copolymer structure on the coating and marine antifouling performances. While hydrophilic poly(SBMA) homopolymer formed a thin coating with ineffective fouling resistance in the presence of marine sediments, the copolymer coatings were much more hydrophobic yet dramatically reduced diatom adhesion even when silts were present. The random copolymer with the optimal composition outperformed the block copolymer analog despite the comparable coating thickness. Our results provide a design principle for the synergetic antifouling mechanism stemming from the hydrophilicity-hydrophobicity balance in the random sequence, where hydrophilic and hydrophobic functionalities are arranged close to each other and exposed on the outermost surface together in contact with water to combat both diatom and silt adsorption. Excellent

antifouling performance on stainless steel substrates combined with long-term stability suggests the potential of the developed materials for practical applications.

Experimental section

Materials

Dopamine hydrochloride (DA, 98%), tris(hydroxymethyl) aminomethane hydrochloride (Trizma HCl, 99%), tris(hydroxymethyl) aminomethane (Trizma base, 99%), dimethyl sulfoxide (DMSO, 99%), 1,1,1,3,3,3-hexafluoro-2-propanol (HFIP), sodium trifluoroacetate (NaTFA), phosphate-buffered saline (PBS), 3-(*N*-2-methacryloyloxyethyl-*N,N*-dimethyl)ammonatopropane-sulfonate (SBMA, >95%), lithium chloride (LiCl, >99.0%), 2,2,2-trifluoroethyl methacrylate (TFEMA, 99%), 4-cyano-4-(dodecylsulfanyltiocarbonyl)sulfanyl pentanoic acid (CDPA, >98%), and 4-cyano-4-(phenyl-carbonothioylthio) pentanoic acid (CTP) were purchased from Sigma-Aldrich (St. Louis, MO). All monomers were passed through a basic alumina column to remove inhibitors before polymerization. 2,2'-Azobis(2-methylpropionitrile) (AIBN, 98%) was purchased from Junsei (Tokyo, Japan) and purified by recrystallization in methanol. 2,2,2-Trifluoroethanol (TFE, >99%) and zirconium(IV) chloride (ZrCl₄) were purchased from Alfa Aesar (Ward Hill, MA) and used as received. The marine benthic diatom *Amphora coffeaeformis* was obtained from the Korea Marine Microalgae Culture Center (KMMCC) and grown in 100 mL of f/2 culture medium at 18 °C for 10 d. Spectra/por membranes purchased from Spectrum Laboratories (Rancho Dominguez, CA) with molecular weight cutoff of 3.5 kg mol^{-1} were used for dialysis.

Synthesis of random poly(SBMA-*r*-TFEMA)

Synthesis of poly(SBMA-*r*-TFEMA) ([SBMA]:[TFEMA] = 1:3) is given as an example. A solution of TFEMA (2.9900 g, 17.77 mmol), AIBN (0.0030 g, 0.018 mmol), CDPA (0.0747 g, 0.185 mmol), and SBMA (1.500 g, 5.37 mmol) was prepared in DMSO (25 mL) with LiCl (0.2 g). The solution was split into 6 ampoules, degassed via three cycles of freeze-pump-thaw, and flame-sealed under vacuum. The ampoules were heated to 70 °C for different time intervals (2, 4, 8, 10, and 120 h), cooled to room temperature, and then opened to terminate the polymerization. The product was recovered by precipitation in a mixture of *n*-hexane/ethanol (1/1 (v/v)) and dried under vacuum. The low-molar mass product obtained by the 2-h polymerization was dialyzed against deionized (DI) water for 3 d and then freeze-dried for 3–5 d.

Other poly(SBMA-*r*-TFEMA)s with different compositions and molar masses were synthesized following the protocol described above by adjusting the feed composition and the polymerization time.

Synthesis of block poly(SBMA-*b*-TFEMA)

A solution of TFEMA (1.51 g, 8.98 mmol), AIBN (0.0067 g, 0.0408 mmol), and CDPA (0.0329 g, 0.0816 mmol) was prepared in TFE (5 mL). The solution was placed into an ampoule, degassed via three cycles of freeze-pump-thaw, and flame-sealed under vacuum. The ampoule was heated to 70 °C for 20 h, cooled to room temperature, and then opened to terminate the polymerization. The solution was dialyzed against methanol for 12 h. The poly(TFEMA) homopolymer was obtained by evaporating the solvent under reduced pressure (1.20 g, 77.8%). To extend the chain with poly(SBMA), poly(TFEMA) (0.500 g, 0.029 mmol), SBMA (0.320 g, 1.15 mmol), and AIBN (0.0024 g, 0.0144 mmol) were dissolved in TFE (2 mL). The solution was transferred into an ampoule and polymerized at 70 °C for 20 h following the protocol described above. The product was obtained by precipitation in a mixture of *n*-hexane/ethanol (1/1 (v/v)) and dried under vacuum.

Synthesis of poly(SBMA)

SBMA (4.470 g, 16.00 mmol), AIBN (0.0016 g, 0.010 mmol), and CTP (0.0056 g, 0.020 mmol) were dissolved in TFE (10 mL). The solution was placed into a Schlenk flask and degassed via four cycles of freeze-pump-thaw. The mixture solution was heated to 70 °C for 24 h under nitrogen atmosphere, cooled to room temperature, and then opened to terminate the polymerization. The product was obtained as a precipitate in methanol, which was filtered off, and again stirred in methanol for 3 h. The final product was dried under vacuum.

Polymer characterization

Molar mass of poly(SBMA) was determined by gel permeation chromatography (GPC) using an RI detector (Shimadzu Corporation RID-20 A, Kyoto, Japan) equipped with a Shodex OHpak SB-804 HQ GPC column (Munich, Germany) and an OHpak SB-G 6 B guard column. An aqueous solution of sodium nitrate (0.1 M, pH 7.4) was used as the eluent, and the flow rate was set to 0.5 mL min⁻¹ at 30 °C. Calibration was performed using poly(ethylene glycol) standard ReadyCal Set (240–217,000 g mol⁻¹). Molar masses of poly(TFEMA), poly(SBMA-*r*-TFEMA), and poly(SBMA-*b*-TFEMA) were determined by Tosoh EcoSEC HLC-8320 GPC using an RI detector (Tokyo, Japan) equipped with two TSKgel SuperAWM-H GPC columns and a TSKgel guard column SuperAW-H. An HFIP solution of NaTFA sodium trifluoroacetic acid (10 mM) was used as the eluent at a flow rate set to 0.3 mL min⁻¹ at 40 °C. Calibration was performed using poly(methyl methacrylate) standards (160–675,500 g mol⁻¹). Nuclear magnetic resonance (NMR) spectroscopy (AVANCE 400 MHz, Bruker, Billerica, MA) was used to determine the TFEMA-to-SBMA ratio of the poly(SBMA-*r*-TFEMA) and poly(SBMA-*b*-

TFEMA) using deuterated methanol and TFE as the solvent. The residual NMR solvent peak was used as an internal reference. The SBMA molar fraction in the copolymers was estimated based on the integral ratio of the peak at 4.3 ppm (CF₃CH₂– and –N⁺CH₂CH₂O–) to that at 2.8 ppm (–CH₂SO₃[–]).

Poly(SBMA) coating on solid substrates

Prior to poly(SBMA) coating, the solid substrates were sonicated in ethanol for 5 min and dried under a stream of nitrogen gas. The solid substrates were then immersed in a DA solution (10 mM, Tris buffer, pH 8.4) for 3 h and washed with DI water. Poly(SBMA) was dissolved in 1× PBS (10 mg/mL, pH 7.4) and dropped onto the pDA-coated substrates, which were then rotated at 3000 rpm for 180 s. The poly(SBMA)-immobilized substrates were immediately immersed in a methanol solution of ZrCl₄ (0.1 M) followed by an aqueous ZrCl₄ solution (0.1 M) for 10 min each. Finally, the substrates were immersed in 1× PBS (pH 7.4) for 1 h, rinsed with DI water, and dried under a stream of nitrogen gas.

Copolymer coating on solid substrates

The solid substrates were sonicated in ethanol for 5 min and dried under a stream of nitrogen gas. The solid substrates were then immersed in a DA solution (10 mM, Tris buffer, pH 8.4) for 3 h and washed with DI water. Amphiphilic copolymer was dissolved in TFE (10 mg/mL) and dropped onto the pDA-coated substrates, which were then rotated at 3000 rpm for 180 s. The copolymer-immobilized substrates were immediately immersed in a methanol solution of ZrCl₄ (0.1 M) for 10 min, followed by immersion in a 1:2 mixture of TFE and ethanol containing ZrCl₄ (0.1 M) for another 10 min. Finally, the substrates were rinsed with DI water and dried under a stream of nitrogen gas.

Coating characterization

X-ray photoelectron spectroscopy (XPS) was conducted using a Thermo Scientific NEXSA G2 spectrometer (Waltham, MA) equipped with an Al K α X-ray source. The coating thickness on the solid substrates was measured using a J. A. Woollam alpha-SE spectroscopic ellipsometer (Lincoln, NE) in the wavelength range of 390–900 nm at an incident angle of 70°. The data were fitted to a Cauchy model using CompleteEASE software (refractive index $n(\lambda) = 1.45 + 0.01/\lambda^2$). Static water contact angle measurements were performed with Surface Electro Optics Phoenix-10 (Suwon, South Korea) to determine the hydrophobicity of the coatings. The water contact angle was typically measured after a liquid drop had been in contact with the surface for 10–15 s. Atomic force microscopy (AFM) images were obtained in the non-contact mode using a Park Systems XE-7 microscope

(Suwon, South Korea). Fluorescence images were obtained using a Nikon LV100ND optical microscope (Tokyo, Japan) equipped with a fluorescence imaging system.

Dynamic marine diatom adhesion test

A suspension of marine diatoms with a $3 \mu\text{g mL}^{-1}$ chlorophyll content was prepared. Natural sand sediment (silt) obtained from the test site at Port Boryeong, South Korea ($36^{\circ}20'15.1''\text{N}$ $126^{\circ}31'57.5''\text{E}$) were added to the marine diatom suspension at a concentration of 37.5 g L^{-1} to mimic the real environments. The resulting suspension was spread onto the solid substrates and incubated in a Daihan Scientific Co. MaXshakeTMRP2 reciprocating shaker (Wonju, South Korea) at 20°C and 50 rpm for 24 h. The solid substrates were gently washed with artificial seawater. The fouling behavior of diatoms on the solid substrates was analyzed using fluorescence microscopy.

Statistical analysis

For the data obtained from at least three samples independently prepared, statistical analysis was carried out using an analysis of variance (ANOVA) followed by Tukey's Test for multiple comparisons (significance level $\alpha = 0.05$). The results are presented as the mean $\pm$ standard deviation.

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Author contributions

J.J. and M.S. contributed equally to this work. S.M.K. and M.S. conceptualized the idea. M.S. performed polymer synthesis and characterization. J.J. prepared polymer coating, evaluated the coating characteristics, and performed the fouling tests. J.J. and S.M.K. wrote the original draft. Every author reviewed and edited the manuscript. S.M.K. and M.S. supervised all aspects of the research.

Competing interests

The authors declare no competing interests.

Ethics approval and consent to participate

This paper is not applicable as it does not report on or involve any animals, humans, human data, human tissue or plants.

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